🏥 Hospital Operations & Financial Performance Intelligence Dashboard

(Power BI)

🌟 Project Overview

This project presents an end-to-end **Healthcare Business Intelligence Dashboard** built using **Power BI** to analyze hospital operations, patient demographics, bed utilization, and financial performance. The objective was to design a structured relational data model, implement robust DAX measures, and build executive-level analytics that connect patient flow, operational efficiency, and revenue performance.

The dashboard integrates **patient management analytics, hospital operational monitoring, resource utilization tracking, and financial performance insights** to simulate a real-world healthcare analytics environment. It provides a **360° operational and financial view** of hospital performance, enabling decision-makers to monitor patient demand, optimize bed utilization, and track revenue drivers.

🎯 Business Objectives

- ★ Measure overall hospital patient and revenue performance
- ★ Monitor hospital operational efficiency and resource utilization
- ★ Track bed occupancy and patient flow trends
- ★ Identify department-level performance and service demand
- ★ Analyze patient demographics and geographic distribution
- ★ Evaluate hospital billing and payment performance
- ★ Provide actionable insights for hospital management and planning

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📂 Dataset Description

The dataset consists of structured hospital system tables representing patient admissions, hospital departments, doctors, billing records, and patient demographics.

Table	Description
Admissions	Patient admission records including admission type, department, doctor, and bed allocation
Patients	Patient demographic information including age, gender, and city
Doctors	Doctor details and department assignments
Departments	Hospital department information
Billing	Billing transactions including total bill amount and payment status
Date	Calendar table used for time intelligence
MyMeasures	Dedicated table to store all DAX measures

The dataset simulates a **real-world hospital data warehouse environment** used for operational and financial analysis.

🧱 Data Model Design

The model follows a clean **star-schema structure** for optimal performance and analytical flexibility.

```
                              dateset
                                 |
                                 |
          Patients ———— Admissions ———— Departments
              |                |                |
              |                |                |
              |              Doctors            |
MyMeasures
              |                |                |                    (Measure
Storage Only)
              |                |                |
              └———————— Billing ————————————————┘
```

Model Characteristics

- One-to-Many relationships maintained
- Single-direction filtering applied across fact tables
- Dimension tables provide consistent filtering context
- No ambiguous relationships or circular dependencies
- Dedicated **MyMeasures table** used for centralized DAX management

Model Architecture

- Admissions acts as the primary operational fact table
- Billing tracks hospital revenue and financial transactions
- Patients provides demographic segmentation
- Doctors enables performance and workload analysis
- Departments supports operational performance evaluation
- Date table enables time-based trend analysis

🧮 Key DAX Measures Implemented

1️⃣ Core Hospital KPIs

- Total Patients
- Total Revenue
- Bed Occupancy Rate
- Active Patients
- Total Admissions
- Total Doctors

2️⃣ Operational Efficiency Metrics

- Average Length of Stay
- Admissions by Department
- Admissions by Doctor
- Department Performance
- Bed Occupancy Trend
- Admission Type Distribution (OPD, IPD, Emergency)

3️⃣ Patient Demographics Analysis

- Patient Age Distribution
- Gender Distribution
- City-wise Patient Count
- Patient Growth Trend
- Total Patients by Demographic Segment

4️⃣ Financial Performance Metrics

- Total Bills
- Average Bill Value
- Revenue by Department
- Average Treatment Cost
- Payment Status Distribution
- Co-payment Distribution

All measures were implemented using proper **filter context handling**, **time intelligence functions**, and **aggregation logic** to ensure KPI accuracy across slicers and filters.

📈 Dashboard Pages

1️⃣ Executive Overview — Hospital Performance

KPIs :

- Total Patients
- Total Revenue
- Bed Occupancy Rate

Visuals :

- Monthly Revenue Trend
- Bed Occupancy Trend
- Department Performance
- Revenue by Admission Type

Interactive Feature :

Button-based navigation to display a detailed operational table showing:

- Doctor
- Department
- Admission Count
- Revenue
- Operational Efficiency

Purpose : Provide a high-level executive overview of hospital operations, including patient demand, revenue growth, and resource utilization.

2️⃣ Operational Analysis — Hospital Efficiency

KPIs :

- Total Doctors
- Average Length of Stay

- Total Admissions

Visuals :

- Admissions by Doctor
- Admissions by Department
- Average Length of Stay by Department
- Admission Type Distribution

Interactive Feature :

Button-based navigation to display a detailed operational table showing:

- Doctor
- Department
- Admission Count
- Revenue
- Operational Efficiency

Purpose : Evaluate hospital operational efficiency, doctor workload distribution, and department-level performance.

3 Patient Analysis — Demographics & Demand

KPIs :

- Total Patients

Visuals :

- Age Distribution (Line & Clustered Column Chart)
- Gender Split (Pie Chart)
- City-wise Patients (Map View)

Interactive Feature :

Button-based navigation to display a detailed operational table showing:

- Doctor
- Department
- Admission Count
- Revenue
- Operational Efficiency

Purpose : Analyze patient demographics and geographic distribution to support hospital planning and service allocation.

4️⃣ Financial Analysis — Revenue & Billing Intelligence

KPIs :

- Average Bill Value
- Total Bills

Visuals :

- Revenue by Department
- Average Treatment Cost by Department
- Payment Status Distribution
- Co-payment Distribution

Interactive Feature :

Button-based navigation to display a detailed operational table showing:

- Doctor
- Department
- Admission Count
- Revenue
- Operational Efficiency

Purpose : Monitor hospital financial performance, billing efficiency, and payment behavior.

📝 Overall Dashboard Objective

To provide a complete **hospital intelligence framework** by integrating:

- Patient demand analysis
- Operational efficiency monitoring
- Resource utilization tracking
- Financial performance measurement

This enables hospital administrators to improve service delivery, optimize resource allocation, and maintain financial sustainability.

🎨 Advanced UX Features

- ★ Interactive page navigation buttons
- ★ Dynamic KPI cards with monthly comparison
- ★ Conditional color indicators for performance trends
- ★ Map-based geographic patient analysis
- ★ Centralized MyMeasures table for DAX management
- ★ Consistent visual theme and layout across pages

The dashboard balances **analytical depth**, **performance**, and **professional user experience design**.

🔍 Key Business Insights

- ★ Bed occupancy trends reveal hospital capacity utilization patterns.
- ★ Certain departments receive significantly higher patient demand.
- ★ Average length of stay varies across departments.
- ★ Emergency admissions contribute a major share of patient volume.
- ★ Revenue performance is closely linked to patient admission trends.
- ★ Payment status distribution highlights billing collection efficiency.

Note: Dataset is synthetic; therefore some patterns may appear smoother than real-world hospital data.

💼 Business Recommendations

- ★ Monitor bed occupancy levels to optimize hospital capacity planning.
- ★ Allocate additional resources to high-demand departments.
- ★ Reduce patient length of stay through process optimization.
- ★ Improve billing efficiency for pending payment cases.
- ★ Track patient demographics to plan future healthcare services.

⚒️ Tools Used

- ★ Power BI
- ★ Data Modeling (Star Schema Design)
- ★ Advanced DAX Measures
- ★ Power Query (Data Transformation)
- ★ Healthcare Operations Analytics
- ★ Business Intelligence Visualization Design

🚀 Portfolio Value

This project demonstrates:

- ★ End-to-end BI dashboard development
- ★ Healthcare operations analytics
- ★ Advanced DAX implementation
- ★ Interactive dashboard design
- ★ Data modeling best practices
- ★ Business storytelling through dashboards

Suitable for Roles :

- ★ Data Analyst
- ★ Business Intelligence Analyst
- ★ Healthcare Data Analyst
- ★ Operations Analyst

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